Spectral Early Warning Signals for Model Collapse in Recursive Synthetic Training

Anonymous ACL submission

Abstract

During the early stages of recursive synthetic training, spectral indicators of the embedding space suggest the model is improving—even as downstream performance is already degrading. We reveal the source of this misleading signal by tracking spectral properties of the embedding matrix across 10 recursive generations of Pythia-410M, with pilot validation on Pythia-1B. The result is a reproducible three-phase pattern, consistent across three random seeds (spectral metric CV $< 1.3\%$ at all generations): spectral decompression (generations 0–2), where effective rank and spectral log-volume (SLV) increase while HellaSwag accuracy drops 8.0%; collapse onset (generations 2–4), where spectral indicators reverse; and accelerating collapse (generations 4–10). Of seven indicators evaluated, SLV is the most effective for automated monitoring: it requires only a single forward pass (< 1 second on one GPU, compared to $\sim$70 minutes for benchmark evaluation), captures volume changes across all spectral directions, and declines monotonically after the decompression peak. By varying the real-data mixing ratio across five levels, we construct a collapse phase diagram that localizes the critical mixing ratio r_{crit} to the interval $(0.25, 0.3]$ across three random seeds, separating monotonic spectral expansion from a transitional regime with initial contraction followed by recovery. A proportional controller demonstrates an inherent signal attenuation: moderate real-data mixing prevents the spectral contraction that would activate the controller, so reactive spectral feedback becomes ineffective precisely when prevention succeeds. Nevertheless, the empirically calibrated r_{crit} provides a practical workaround: setting the mixing ratio above this threshold prevents collapse without real-time monitoring, reducing the problem to a one-time offline calibration requiring only forward-pass spectral checks.

1 Introduction

Synthetic data has become integral to language model training. Instruction tuning, knowledge distillation, and data augmentation all routinely use model-generated text, and as synthetic corpora are recycled into training pipelines, the process often becomes recursive: each generation of models is fine-tuned on outputs from the previous one. Recent work demonstrates that such recursive synthetic training triggers *model collapse*, a progressive erosion of distributional tails and downstream capability (Shumailov et al., 2024; Dohmatob et al., 2024b; Alemohammad et al., 2024). The problem is especially pressing as web-scraped corpora increasingly contain model-generated content, a trend that introduces implicit recursive loops even without deliberate data reuse.

While model collapse is now well-characterized through theoretical analysis (Dohmatob et al., 2024b) and empirical studies (Shumailov et al., 2024; Alemohammad et al., 2024), its effect on the geometric structure of learned representations remains poorly understood. Characterizing how the embedding space contracts during recursive training could yield computationally cheap collapse indicators from internal model states alone, without text generation or benchmark evaluation.

Current collapse detection depends on downstream metrics such as perplexity and benchmark accuracy, or on analysis of generated text through lexical diversity measures. Both approaches require either costly inference or external evaluation infrastructure at every generation. SIGMA (Gu et al., 2026) introduced spectral monitoring through embedding covariance analysis and showed that spectral indicators can track collapse, but validated only on OPT-125M over short generation horizons. Four gaps remain open. First, no work has systematically characterized spectral dynamics beyond a few generations at 410M+ parameters. Second, the

temporal relationship between spectral indicators and downstream degradation has not been examined. Third, no study has connected spectral indicators to closed-loop collapse prevention through the lens of early warning signals (EWS) (Scheffer et al., 2009). Fourth, the relationship between real-data mixing ratios and spectral collapse has not been mapped: it is unknown how much real data suffices to prevent spectral contraction, or whether a sharp phase boundary exists.

We address these gaps by tracking spectral indicators of the embedding matrix across 10 recursive generations of Pythia-410M-deduped (Biderman et al., 2023), with pilot validation on Pythia-1B-deduped. Our analysis reveals a reproducible three-phase pattern (Figure 1): spectral decompression (Gen 0–2), where effective rank (r_{eff}) and spectral log-volume (SLV) rise while downstream performance already declines; collapse onset (Gen 2–4), where spectral indicators reverse; and accelerating collapse (Gen 4–10), where contraction accelerates. Among these indicators, SLV is the most effective for automated monitoring: it requires only a forward pass on held-out data (no text generation), captures distributed volume changes across all spectral directions, and declines monotonically after its peak. We further design a proportional controller and demonstrate an inherent signal attenuation: moderate real-data mixing prevents the spectral contraction that would trigger the controller, so reactive spectral feedback becomes ineffective precisely when collapse prevention succeeds.

Our contributions are:

- **C1:** The first multi-seed spectral-downstream characterization at 410M+ scale, revealing dynamics invisible in prior short-horizon studies and establishing a reproducible three-phase collapse pattern (decompression, onset, acceleration) across 10 generations (§3.2, Table 1).
- **C2:** SLV as a computationally cheap collapse monitor with monotonic post-peak decline and a HellaSwag-spectral asymmetry: downstream degradation begins during the spectral decompression phase, demonstrating that spectral expansion does not indicate model health (§3.2).
- **C3:** Quantitative characterization of signal attenuation under mixing: the spectral controller signal degrades continuously as the real-data fraction approaches r_{crit} and vanishes entirely above it, a limitation robust to threshold reduction (§2.2, §3.3).

Supporting these findings, we construct a spectral collapse phase diagram across multiple mixing ratios that localizes r_{crit} empirically, with the bound robust to both the EMA smoothing parameter and random seed (§2.3, §3.3, §3.5).

2 Method

2.1 Spectral Metrics

Given a language model at generation t, we extract last-layer embeddings for n=1,000 held-out samples to form $\mathbf{E} \in \mathbb{R}^{n \times d}$. Let $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_{\min(n,d)}$ denote the singular values of $\mathbf{E}$. We track three complementary spectral indicators:

Spectral Log-Volume (SLV). SLV $= \sum_{i:\,\sigma_i > \epsilon} \log \sigma_i$, where $\epsilon = 10^{-10}$ excludes numerically zero components. When $n < d$ (as in our setting with n=1,000, d=1,024), $\mathbf{E}$ has at most n non-zero singular values; SLV thus measures the log-volume of the occupied subspace rather than the full d-dimensional space. Because it sums over all non-negligible singular values, SLV is sensitive to contraction along any spectral direction, not only the dominant ones.

Effective Rank. $r_{\text{eff}} = \exp(H(\mathbf{p}))$, where $p_i = \sigma_i / \sum_j \sigma_j$ and H is the Shannon entropy, measuring how many dimensions carry substantial variance (uniform $\to d$, concentrated $\to 1$).

Leading Singular Value. σ_1 tracks the dominant spectral direction; rising σ_1 under falling r_{eff} signals representational collapse into a low-dimensional subspace.

2.2 Adaptive Mixing Controller

We propose a closed-loop controller that adjusts the real-data mixing ratio r_t at each generation based on the rate of spectral change. The update rule is:

$$r_{t+1} = \text{clip}\big(\tfrac{B_{\text{rem}}}{T-t} + k \cdot \max(0,\, \Delta_t - \tau),\; r_{\min},\; r_{\max}\big) \quad (1)$$

where B_{rem} is the remaining real-data budget (fraction of total budget not yet consumed), T is the total number of generations, τ=0.002 is a sensitivity threshold, and k=10 is the controller gain. The signal Δ_t is the baseline-normalized SLV drop: $\Delta_t = (\widetilde{S}_{t-1} - \widetilde{S}_t)/|\widetilde{S}_0|$, where $\widetilde{S}_t$ is the exponential moving average (EMA)-smoothed SLV (α=0.5) and $\widetilde{S}_0$ is initialized from the pretrained

model (Gen 0, SLV=2366.68). The signed form ensures that spectral decompression ($\Delta_t < 0$) does not trigger the controller; only contraction activates the $\max(0, \cdot)$ gate in Eq. 1. The output is clipped to $[r_{\min}, r_{\max}] = [0.20, 0.80]$ and subject to $\sum_t r_t \leq B_{\text{total}}$. The lower bound ensures a minimum synthetic fraction; $r_{\max}$ defines the controller's operational range.

The first term distributes the remaining budget uniformly across future generations; the model degrades gracefully even without spectral signal. The second term injects additional real data when SLV contraction exceeds the threshold τ, proportional to the severity of the spectral drop. This design requires no lookahead and operates with a single hyperparameter choice (k) beyond the budget constraint.

Controller Silence Condition. Define r_{crit} as the minimum constant mixing ratio such that the EMA-smoothed SLV is non-decreasing from the second generation onward: $\widetilde{S}_t \geq \widetilde{S}_{t-1}$ for all $t \in \{2, \ldots, T\}$ (excluding the initial fine-tuning transient at t=1). For conditions with multiple seeds, the condition is evaluated on the cross-seed mean trajectory. When a constant ratio $r \geq r_{\text{crit}}$ is applied, $\Delta_t \leq 0 < \tau$ for all $t \geq 2$ and any $\tau > 0$, so the controller boost is identically zero. This τ-independence means the signal attenuation cannot be resolved by lowering the sensitivity threshold; the same argument extends to cumulative variants. Our experiments (§3.3) empirically bound $0.25 < r_{\text{crit}} \leq 0.3$ over Gen 2–10 for Pythia-410M on OpenWebText.

Second-Order Extension. A natural extension monitors the rate of change of the controller signal itself:

$$\Delta_t^2 = \widetilde{\Delta}_{t-1} - \widetilde{\Delta}_t \tag{2}$$

where $\widetilde{\Delta}_t$ is the EMA-smoothed first-order signal (α_2=0.3). A sustained negative Δ_t^2 indicates decelerating SLV growth, a potential precursor to stagnation even when SLV is still rising. We evaluate this indicator in §3.5.

2.3 Collapse Phase Diagram

We define the *collapse phase diagram* as the mapping from constant mixing ratio r to the SLV trajectory over T generations. For each r, we classify the outcome: *collapse* if SLV declines monotonically after the decompression phase (Gen 2 onward), *stable* if SLV is non-decreasing, and *transitional* if SLV shows non-monotonic behavior after Gen 2 (both increases and decreases within the observation window). The critical mixing ratio r_{crit} (§2.2) is the boundary between collapse and stability in this diagram. We estimate r_{crit} by evaluating multiple constant ratios and identifying the lowest r for which $\widetilde{S}_t \geq \widetilde{S}_{t-1}$ holds for all $t \in \{2, \ldots, T\}$.

Gen	r_{eff}	SLV	PPL	HeSw	D-4	σ_1
0	417.55	2366.68	9.55	.413	.738	1942.8
2	**496.17**	**2467.98**	13.56	.380	.505	718.7
5	479.77	2365.99	24.50	.341	.412	749.2
7	464.10	2293.24	37.82	.319	.375	788.4
10	438.05	2181.35	70.99	.296	.324	874.6

Table 1: Spectral and downstream metrics at selected generations (Pythia-410M, mean of three seeds 42, 43, 44; sample s.d. at Gen 10: r_{eff} ±4.8, SLV ±28.3, PPL ±5.6, HeSw ±.006, D-4 ±.15). Bold marks the spectral peak.

3 Experiments

3.1 Experimental Setup

We use the publicly released final Pythia-410M-deduped checkpoint (Biderman et al., 2023) as our base model and fine-tune on a 50K-subset of OpenWebText. Starting from the pretrained checkpoint (generation 0), each subsequent generation fine-tunes on synthetic text produced by the previous generation, for a total of 10 recursive generations. At each generation we compute the spectral indicators from §2.1 on 1K held-out last-layer embeddings, paired with three downstream metrics: perplexity, HellaSwag accuracy (0-shot), and distinct-4. These three metrics (language modeling capability, natural language understanding, and generation diversity) cover the primary evaluation dimensions accessible to a 410M-parameter model.

For the mixing comparison (§3.3), we evaluate three strategies sharing the same total real-data budget: (1) **pure collapse** (r_t=0), (2) **fixed mix** (r_t=0.50), and (3) **adaptive** (k=10, Eq. 1). To construct the collapse phase diagram (§3.5), we additionally evaluate fixed mixing at $r \in \{0.2, 0.25, 0.3\}$.

3.2 Spectral Collapse Characterization

Table 1 reports spectral and downstream metrics across 10 recursive generations, which reveal a three-phase structure.

Phase I: Spectral Decompression (Gen 0–2). In the first two generations, r_{eff} rises from 417.6 to

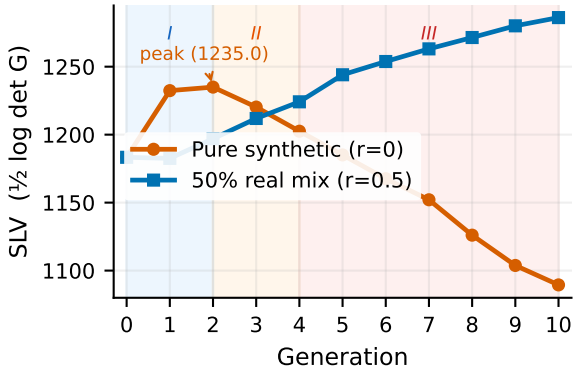


Figure 1: SLV across 10 recursive generations of Pythia-410M (solid: r=0, 3-seed mean; dashed: r=0.5, single seed) under pure-synthetic training and 50% real-data mixing. Three phases are visible in the pure-synthetic trajectory: decompression (Gen 0–2, SLV peaks at 2468), onset (Gen 2–4), and acceleration (Gen 4–10, SLV falls 11% from peak to 2181). Under 50% mixing, SLV rises monotonically (except a negligible Gen 1 dip) to 2573 at Gen 10, 8.7% above baseline and 105 points above the pure-collapse peak.

496.2 (+18.8%) and SLV increases from 2366.7 to 2468.0 (+4.3%), while σ_1 drops sharply from 1942.8 to 718.7 as spectral mass redistributes from the dominant direction to smaller components. The consistency of the peak position (Gen 2 across all three seeds) and its coincidence with the qualitative reversal of σ_1 dynamics (from declining to rising) suggest that the phase boundary reflects a structural transition rather than a statistical artifact. Crucially, this decompression does not indicate improved model quality: HellaSwag drops from 0.413 to 0.380 (-8.0%) and distinct-4 falls from 0.738 to 0.505 (-31.6%) during the same interval. HellaSwag already declines 5.4% from Gen 0 to Gen 1, precisely when spectral indicators are still rising; this asymmetry suggests that decompression captures distributional broadening rather than preservation of task-supporting representations. A monitoring system relying solely on spectral metrics during Phase I would incorrectly conclude the model remains healthy; joint spectral-downstream tracking (Gu et al., 2026) is advisable. SLV's value lies not in temporal precedence but in computational efficiency (<1 s vs. ∼70 min for benchmarks) and cross-seed robustness (CV $< 1.3\%$).

Phase II: Collapse Onset (Gen 2–4). Both r_{eff} and SLV reverse after Gen 2 (r_{eff}: 496.2→493.6, SLV: 2468→2440), and this reversal marks the transition from decompression to collapse while perplexity continues its monotonic rise.

Phase III: Accelerating Collapse (Gen 4–10). Spectral contraction accelerates: the mean per-generation r_{eff} decline in Phase III is $1.78 \pm 0.22\%$/gen (mean ± s.d. across three seeds). By Gen 10, r_{eff} falls to 438.1 (-11.7% from peak), SLV drops to 2181.4 (-7.8% from baseline), perplexity reaches 71.0 ($+643\%$), and σ_1 reverses its earlier decline, reaching 874.6 at Gen 10, a sign of renewed spectral concentration (Figure 1).

Cross-Scale Pilot. Repeating the protocol on Pythia-1B (Gen 0–3) confirms the same decompression-onset structure (r_{eff} peak at Gen 2, SLV peak one generation earlier). SLV's low cost and monotonic post-peak decline make it well-suited as an automated collapse monitor. In our setup, SLV computation (1,000-sample forward pass plus SVD of the $n \times d$ embedding matrix) takes < 1 second on a single GPU, whereas HellaSwag 0-shot evaluation requires ∼70 minutes end-to-end under identical hardware.

3.3 Adaptive Mixing Comparison

Under pure synthetic training (§3.2), SLV peaks at Gen 2 (2468) and declines monotonically; with 50% mixing, SLV rises to 2573 at Gen 10 (105 points above the pure-collapse peak), with only a negligible Gen 1 dip ($\Delta_1 \approx 0.0003$, well below τ).

The adaptive controller (k=10) remains silent through all ten generations: $\max \Delta_t = 0.0003$ at Gen 1, well below τ=0.002, and $\Delta_t \leq 0$ for all $t \geq 2$. Since the controller monitors the *rate* of spectral change, even moderate mixing that does not prevent downstream degradation (PPL +45% at r=0.2) suffices to suppress the signal. Pure collapse produces SLV decline from Gen 3 while r=0.5 yields sustained growth, bounding $0.2 < r_{\text{crit}} \leq 0.5$ (§3.5).

Despite sustained spectral expansion, downstream metrics degrade steadily: perplexity increases 59% (9.55 → 15.20) and distinct-4 falls 87% (.74 → .095) over ten generations, the lowest D-4 of any condition, below even pure collapse (r=0: 0.324). This reinforces the asymmetry from §3.2: spectral health under mixed training does not guarantee model health, and signal attenuation renders per-generation reactive controllers ineffective precisely when collapse prevention succeeds. The relationship between r_{crit} and signal suppression is inherent to any rate-based indicator; our empirical contribution is demonstrating that this creates a

Indicator	Source	Peak	Mono?	Rel Δ	Rev.?
SLV	emb-SVD	Gen 2	✓	11.8%	✓
RankMe	emb-SVD	Gen 2	✓	4.7%	✓
Token ent.	logit	—	✓	9.6%	✓
Stable rank	attn wt.	—	✓	14.0%	✓
Spec. norm	attn wt.	—	✓	8.0%	✓
Isotropy	emb	—	× (6)	75%	✓
PPL growth	ppl	—	∼ (1)	—	✓

Table 2: Alternative early warning indicators under pure-synthetic training (r=0, Pythia-410M, 10 gen). **Peak**: generation at which the indicator reverses (— = monotonic from Gen 0). **Mono?**: post-peak monotonicity (✓ = yes; number = violations). **Rel Δ**: $|v_{10} - v_0|/|v_0|$, magnitude of total relative change Gen 0–10. **Rev.?**: direction reversal under r=0.5 mixing (✓ = reversed).

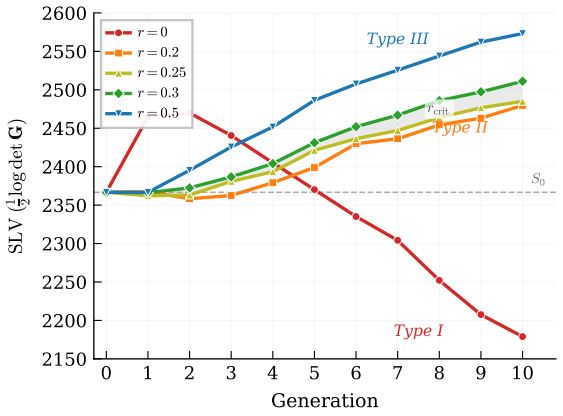


Figure 2: SLV trajectories across mixing ratios (Pythia-410M; r=0 and r=0.25: 3-seed mean; r=0.3: 3-seed mean; r=0.2 and r=0.5: single seed). Type I (r=0): decompression then monotonic collapse. Type II (r=0.2, r=0.25): transient dip followed by recovery. Type III (r=0.3, r=0.5): monotonic expansion from Gen 2 onward. The shaded band marks the critical threshold $r_{\text{crit}} \in (0.25, 0.3]$. The dashed horizontal line indicates the Gen 0 baseline.

continuous, quantifiable degradation of monitoring capability across the sub-critical regime, a phenomenon not previously quantified. This raises the question: is signal attenuation specific to SLV, or does it affect all candidate EWS indicators?

3.4 Alternative EWS Comparison

We compare SLV against six alternative collapse indicators from three signal sources: embedding geometry (RankMe, isotropy), attention weights (stable rank, spectral norm), and output distributions (token entropy, perplexity growth rate).

Table 2 summarizes the comparison under pure-synthetic training (r=0). Two indicators share SLV's decompression-then-collapse trajectory: RankMe (Garrido et al., 2023), the exponential of the normalized singular-value entropy, peaks at Gen 2 and declines monotonically thereafter, though with 2.5× lower signal-to-noise ratio (4.7% relative change vs. 11.8% for SLV). Weight-level and output-level indicators (stable rank, spectral norm, token entropy) change monotonically from Gen 0 with no decompression phase; they cannot distinguish collapse onset from the initial fine-tuning transient. Isotropy is the least reliable indicator: random sampling without a fixed seed introduces approximately 5% variance, and it exhibits six monotonicity violations across ten generations, with direction reversals at Gen 4+ that produce a misleading recovery signal.

The final column of Table 2 reveals that all seven indicators reverse direction under 50% real-data mixing. This universality confirms that signal attenuation (§2.2) is not specific to SLV: any per-generation reactive controller based on these indicators will fall silent when mixing exceeds r_{crit}. Among embedding-level metrics, SLV offers the highest signal-to-noise ratio (2.5× that of RankMe) and, together with RankMe, is the only indicator that exhibits a decompression baseline to distinguish collapse onset from training transients. SLV is the only indicator that combines decompression-phase sensitivity, post-peak monotonicity, and sub-second computation cost; it shares a metric foundation with SIGMA's $\log\det G$ (= 2 × SLV; §4).

3.5 Collapse Phase Diagram

We now map SLV trajectories across mixing ratios to construct the collapse phase diagram defined in §2.3. Figure 2 plots SLV across 10 recursive generations for five mixing ratios; three qualitatively distinct trajectory types emerge:

Type I: Pure Collapse (r=0). SLV exhibits large decompression (+97 points at Gen 1, peaking at Gen 2) followed by accelerating contraction (§3.2).

Type II: Near-critical Mixing (r=0.2). Twenty percent real data suppresses the SLV decompression phase: SLV rises only 0.78 points from Gen 0 to Gen 1 (vs. +98 under r=0), dips at Gen 2 (2367.5 → 2358.3, −9.2). Recovery begins at Gen 3, with per-generation gains accelerating from +4.2 to +30.9 (Gen 6) before moderating, reaching 2479.7 at Gen 10 (+4.8% above baseline). Because $\widetilde{S}_2 = 2362.7 < \widetilde{S}_1 = 2367.1$ (EMA-

smoothed SLV; §2.2), the non-decreasing condition is violated and r=0.2 formally lies below r_{crit}. The trajectory is consistent with the transitional regime (§2.3): initial contraction followed by sustained recovery over the full 10-generation horizon. The smoothed signal never exceeds the controller threshold ($\max \Delta_t = 0.0019$ at Gen 2, below $\tau = 0.002$), so as little as 20% real data silences the controller entirely. The effective rank still rises at Gen 2 (476.7 → 481.6) while the EMA-smoothed SLV registers a net decline ($\widetilde{S}_2 < \widetilde{S}_1$), evidence that smoothed SLV detects spectral contraction earlier at this mixing ratio.

At r=0.25, the full 10-generation trajectory follows the same Type II pattern: SLV dips from Gen 0 to Gen 1 (2366.7 → 2362.1, −4.6; seed 42, representative trajectory), reaches a minimum at Gen 2 (2363.1), then rises monotonically through Gen 10 (three-seed mean 2488.1, +5.1% above baseline). The EMA-smoothed sequence confirms the sub-critical classification: $\widetilde{S}_1 = 2364.4$, $\widetilde{S}_2 = 2363.8$, satisfying $\widetilde{S}_2 < \widetilde{S}_1$ and violating the non-decreasing condition. Controller silence is more pronounced than at r=0.2: $\max \Delta_t = 0.00028 < \tau = 0.002$, a 6.7× weaker signal than the r=0.2 maximum (0.0019), consistent with further signal attenuation as the real-data fraction increases. The amplitude of the EMA violation at r=0.25 varies substantially across seeds ($\widetilde{S}_2 - \widetilde{S}_1 = -0.6, -1.7, -0.9$), consistent with the elevated fluctuations expected near a phase boundary, while trajectories at r=0 and $r \geq 0.3$ show low cross-seed variance ($< 0.5\%$).

Type III: Super-critical Mixing ($r \geq 0.3$). SLV rises monotonically from Gen 2 onward, reaching 2573 at Gen 10 (+8.7% above baseline), consistent with r=0.5 ≥ r_{crit} (§3.3). Per-generation gains remain positive through all post-transient generations ($\Delta_t \in [+11, +35]$ for $t \geq 2$; mean +23.0), sustaining expansion with no reversal toward the contraction trajectory of Type I within the observed horizon.

These three types partition the mixing-ratio axis into sub-critical ($r < r_{\text{crit}}$), transitional, and stable ($r \geq r_{\text{crit}}$) zones. Type classification is based solely on spectral trajectory shape; downstream metrics such as D-4 may diverge, as demonstrated by the non-monotonic D-4 pattern across mixing ratios (Table 3). At r=0.3, SLV expands monotonically from Gen 2 to 2508.2 at Gen 10 (+6.0% above baseline), with sustained positive growth across all eight post-transient generations. This confirms r=0.3 ≥ r_{crit} over the full 10-generation horizon, tightening the bound to $0.25 < r_{\text{crit}} \leq 0.3$. Across three seeds, one individual EMA-smoothed trajectory shows a transient violation of 0.83 points at Gen 2; the remaining two satisfy non-decreasing from $t \geq 2$. The cross-seed mean trajectory is non-decreasing at all post-transient generations (Gen 10 SLV CV = 0.12%), confirming the Type III classification. The critical threshold is robust to the smoothing parameter: recomputing the EMA-smoothed SLV for $\alpha \in \{0.3, 0.5, 0.7\}$ yields $r_{\text{crit}} \in (0.25, 0.3]$ in all cases, with the non-decreasing condition violated at Gen 2 for $r \leq 0.25$ and satisfied on the cross-seed mean for $r \geq 0.3$ regardless of α.

The largest single-generation SLV drop in the first two generations shrinks with r: −9.2 at r=0.2 (Gen 1→2), −4.6 at r=0.25 (Gen 0→1; seed 42), and −0.6 at r=0.3 (Gen 0→1), consistent with a continuous crossover rather than a sharp boundary. The Type II trajectory clarifies what r_{crit} marks in practice: not a boundary between collapse and stability in final outcome, but a transition in trajectory shape. Below r_{crit}, real-data mixing fails to suppress the initial SLV contraction transient but sustains spectral expansion through Gen 10 once the transient passes.

Output Diversity Divergence. At r=0.2, D-4 exhibits a V-shaped recovery (0.438 at Gen 5 → 0.520 at Gen 10), but this pattern does not generalize: at r=0.25, D-4 declines monotonically to 0.342 despite comparable SLV recovery, indicating that spectral expansion does not guarantee diversity restoration. More broadly, D-4 exhibits a non-monotonic relationship with r: it peaks at r=0.2 (.520), exceeds the pure-collapse baseline (.324) at $r \leq 0.25$ but drops below it at $r \geq 0.3$, reaching .095 at r=0.5, the lowest across all conditions (Table 3). This pattern is consistent with increasing real-data anchoring: at high r, the model converges toward the fixed real-data distribution, producing coherent but stylistically homogeneous text. This divergence between spectral health (monotonically improving with r) and n-gram diversity underscores that SLV and D-4 capture orthogonal aspects of model behavior. D-4 exhibits substantially higher cross-seed variance than spectral metrics: at r=0 (s.d. = 0.15 vs. SLV s.d. = 28, a >30× lower coefficient of variation for SLV) and r=0.3 (s.d. = 0.155 vs. SLV s.d. = 2.9; Ta-

r	Type	SLV	r_{eff}	PPL	D-4	$\max \Delta_t$	Gen.
0	I	2181.4	438.1	71.0	.324	—	10
0.20	II	2479.7	511.3	13.83	.520	.0019	10
0.25	II	2488.1	513.8	13.41	.342	.0003	10
0.30	III	2508.2	521.1	13.35	.295	0	10
0.50	III	2573.2	554.5	15.20	.095	0	10

Table 3: Final-generation spectral and downstream metrics across mixing ratios (all runs to Gen 10). $\max \Delta_t$ computed for $t \geq 2$, excluding the initial fine-tuning transient. Type classification follows §2.3. r=0 reports the mean of three seeds (42, 43, 44); r=0.3 the mean of three seeds (42, 43, 44); r=0.2 and r=0.5 report a single seed; r=0.25 the mean of three seeds (42, 43, 44). D-4 exhibits high cross-seed variance at r=0 (s.d. = 0.15) and r=0.3 (s.d. = 0.155).

ble 3), suggesting that n-gram diversity reflects stochastic generation choices rather than the deterministic geometric structure captured by SLV. This pattern generalizes across mixing ratios: at r=0.25, SLV cross-seed CV is 0.12% versus 3.9% for D-4, confirming that spectral geometry captures a deterministic structural property largely independent of stochastic training noise.

Table 3 consolidates the final-generation metrics across all five mixing ratios.

Second-order and alternative triggers. A second-order acceleration indicator Δ_t^2 (Eq. 2) does not discriminate between sub- and supercritical regimes: all trajectories exhibit negative Δ_t^2 in late generations regardless of mixing ratio. A level-based trigger keyed to baseline SLV could detect the transient dip in Type II trajectories, but would not activate under Type III where SLV exceeds baseline at all generations. Finer resolution between 0.25 and 0.3 would narrow the r_{crit} bound but would not change the practical recommendation to set r above 0.3. For practitioners, the phase diagram provides a direct prescription: in our setting, setting $r > 0.3$ prevents spectral collapse without per-generation monitoring.

4 Related Work

Model Collapse. Shumailov et al. (2024) showed that recursive training on model-generated data progressively erodes the tails of learned distributions, a phenomenon they termed *model collapse*. Scaling laws predict tail erosion as a function of recursive depth (Dohmatob et al., 2024b,a); the self-consuming loop accelerates collapse even under data accumulation (Alemohammad et al., 2024), with as little as 1% synthetic contamination sufficing (Dohmatob et al., 2025), though preserving all prior real data can delay it (Gerstgrasser et al., 2024). We complement these output/benchmark-based diagnoses by tracking collapse through the spectral geometry of internal representations, requiring no text generation. Unlike neural collapse (Papyan et al., 2020), which describes last-layer features converging to a simplex ETF in classification, spectral collapse in recursive training reflects progressive embedding volume loss unrelated to class structure.

Spectral Analysis of Representations. Random matrix theory reveals implicit regularization structure in neural network eigenspectra (Martin and Mahoney, 2021); effective rank (Roy and Vetterli, 2007) and RankMe (Garrido et al., 2023) quantify representational dimensionality. The spectral decompression phase, in which singular value spread decreases while effective rank increases, resembles the effect of isotropy regularization (Ethayarajh, 2019); recursive fine-tuning may initially reduce the anisotropy documented in standard training (Gao et al., 2019) before amplifying it during collapse. Recent work has connected spectral properties to training pathologies: He et al. (2025) linked spectral collapse in hidden representations to plasticity loss in continual learning, and Naderi et al. (2025) identified spectral rank collapse in attention matrices as a failure mode of deep transformers. Most directly relevant, Gu et al. (2026) proposed SIGMA, a spectral monitoring framework that tracks singular-value distributions during recursive training on OPT-125M over three generations. Mathematically, our SLV is equivalent to half of SIGMA's $\log\det G$ (both sum log singular values of the embedding matrix), establishing a shared metric foundation. When $n < d$, both metrics face the same rank-deficiency issue; our ϵ-thresholding (§2.1) makes the equivalence approximate rather than exact. Our work extends SIGMA in *scale* (Pythia-410M/1B, 3–8× larger), *temporal depth* (10 generations, revealing a three-phase trajectory invisible at SIGMA's 3-generation horizon), and *methodology* (closed-loop SLV-based mixing control; §2.2). At three generations spectral indicators are still rising (Figure 1), which could lead to the premature conclusion that spectral health implies model health. Finally, *signal attenuation*, whereby successful mixing suppresses spectral indicators that would have warned of collapse, is a

phenomenon that short-horizon monitoring does not encounter. That SIGMA observed analogous spectral shifts on OPT-125M provides preliminary cross-architecture evidence for the generality of spectral collapse dynamics.

Early Warning Signals and Critical Transitions. Early warning signals (EWS) for critical transitions originate in ecology, where rising autocorrelation and variance precede regime shifts (Scheffer et al., 2009, 2012), with detection methods ranging from statistical tests (Dakos et al., 2012) to deep learning (Bury et al., 2021). A central tension is that stabilization interventions suppress the very precursors they rely on (Scheffer et al., 2009, 2012). Rate-induced critical transitions (Siteur et al., 2016) further demonstrate that interventions can themselves alter system dynamics. We encounter a parallel tension: real-data mixing reverses the spectral trajectory entirely (§2.2), unlike ecological stabilization which preserves signal direction; our contribution is quantifying how rapidly this signal degrades across mixing ratios. He et al. (2025) provides a bridge between these domains by connecting spectral collapse to critical transitions in continual learning.

Data Mixing for LLM Training. Data composition significantly affects LM quality (Longpre et al., 2024; Muennighoff et al., 2023), and several methods optimize domain mixtures through proxy models or predictive formulas (Xie et al., 2023; Ye et al., 2025; Chen et al., 2025). These approaches evaluate mixing through loss or benchmarks. Our work reveals a complementary dimension: real-vs-synthetic mixing reshapes the spectral geometry of internal representations, producing a phase diagram (§3.5) that loss-based metrics do not capture, and our controller targets recursive spectral collapse rather than general domain balance.

5 Conclusion

Tracking embedding spectral properties across 10 recursive generations of Pythia-410M reveals that model collapse follows a structured three-phase geometric trajectory, with the decompression-to-onset structure confirmed at 1B; full 10-generation validation at this scale remains future work. Of the indicators we examined, SLV offers the most effective monitoring signal: sub-second computation via a single forward pass, monotonic post-peak decline, and sensitivity to distributed spectral contraction before it manifests as reduced effective rank. Crucially, downstream degradation begins during the decompression phase, demonstrating that spectral expansion does not indicate model health. A collapse phase diagram across five mixing ratios localizes $r_{\text{crit}} \in (0.25, 0.3]$: above this threshold SLV expands monotonically (Type III), while just below it a transitional regime (Type II) exhibits an initial contraction transient followed by recovery.

A proportional controller exposes inherent signal attenuation: the controller signal vanishes at $r \geq 0.3$, and because all seven EWS indicators reverse under sufficient mixing (§3.4), any reactive controller faces this prevention–detectability tradeoff (Scheffer et al., 2009). Nevertheless, spectral indicators reveal whether the current ratio lies in the transitional or stable zone without waiting for downstream degradation.

Although reactive controllers fail under signal attenuation, calibrating r_{crit} once and setting $r > r_{\text{crit}}$ as a static mixing floor reduces collapse prevention to a one-time offline calibration via forward-pass spectral checks. However, spectral stability does not guarantee downstream quality: even at $r = 0.5$, perplexity increases 59% and D-4 drops 87% (Table 3); quality degradation at fixed r is nonetheless gradual and predictable, unlike the accelerating degradation under spectral collapse.

Signal attenuation instantiates the classical EWS intervention–loss tradeoff (Scheffer et al., 2009) via a distinct mechanism—trajectory inversion rather than declining signal-to-noise ratio—and may extend to implicit-loop settings where web corpora contain model-generated content; validating this generalization is future work.

6 Limitations

Scale and Replication. Multi-seed replication covers pure collapse (three seeds), r=0.25 (three seeds), and r=0.3 (three seeds); $r \in \{0.2, 0.5\}$ each report a single seed. SLV is computed from n=1,000 held-out embeddings; bootstrap resampling yields a coefficient of variation below 0.2% at this sample size, and CV remains below 0.4% across $n \in \{250, 5000\}$. We validate at two model scales only (Pythia-410M and a 1B pilot through Gen 3); while SIGMA observed analogous spectral shifts on OPT-125M (Gu et al., 2026), systematic cross-architecture validation remains future work. The 1B pilot covers only three generations, which is sufficient to confirm the decompression peak

but insufficient to characterize the full acceleration phase or to estimate r_{crit} at that scale.

Domain and Horizon. All fine-tuning uses a single data domain (OpenWebText, English web text). Spectral collapse dynamics may differ substantially for code, multilingual, or domain-specific corpora, where the structure of the data manifold and the degree of distributional shift between real and synthetic text vary. Ten recursive generations may be shorter than production pipelines, and the spectral trajectory beyond this horizon is unknown. It is possible that trajectories currently classified as Type II (transitional) eventually revert to collapse at longer horizons, which would shift the effective r_{crit} upward.

Controller Design. The adaptive controller uses a simple proportional design with a single gain parameter k and a fixed threshold τ, without integral or derivative terms. A more sophisticated controller incorporating cumulative drift or rate-of-change signals might partially mitigate signal attenuation, but exploring such designs is beyond the scope of this work.

Training and Compute Details. Fine-tuning uses AdamW (lr 5×10^{-5}, weight decay 0.01, linear warmup 5%, effective batch size 32) for one epoch per generation, with gradient clipping at max norm 1.0. Synthetic text is generated with temperature 1.0, top-k=50, max length 512. All experiments run on NVIDIA RTX PRO 6000 GPUs (96 GB); each 10-generation pipeline requires 15–26 single-GPU hours depending on the mixing ratio, totaling approximately 220 GPU-hours across all reported experiments.

Code and Data. Code and data will be released upon acceptance.

References

Sina Alemohammad, Josue Casco-Rodriguez, Lorenzo Luzi, Ahmed Imtiaz Humayun, Hossein Babaei, Daniel LeJeune, Ali Siahkoohi, and Richard G. Baraniuk. 2024. Self-consuming generative models go MAD. In *The Twelfth International Conference on Learning Representations (ICLR)*.

Stella Biderman, Hailey Schoelkopf, Quentin Anthony, Herbie Bradley, Kyle O'Brien, Eric Hallahan, Mohammad Aflah Khan, Shivanshu Purohit, USVSN Sai Prashanth, Edward Raff, Aviya Skowron, Lintang Sutawika, and Oskar van der Wal. 2023. Pythia: A suite for analyzing large language models across training and scaling. In *Proceedings of the 40th International Conference on Machine Learning (ICML)*.

Thomas M. Bury, R. I. Sujith, Induja Pavithran, Marten Scheffer, Timothy M. Lenton, Madhur Anand, and Chris T. Bauch. 2021. Deep learning for early warning signals of tipping points. *Proceedings of the National Academy of Sciences*, 118(39):e2106140118.

Mayee F. Chen, Michael Y. Hu, Nicholas Lourie, Kyunghyun Cho, and Christopher Ré. 2025. Aioli: A unified optimization framework for language model data mixing. In *The Thirteenth International Conference on Learning Representations (ICLR)*.

Vasilis Dakos, Stephen R. Carpenter, William A. Brock, Aaron M. Ellison, Vishwesha Guttal, Anthony R. Ives, Saran Kefi, Vasilis Livina, David A. Seekell, Egbert H. van Nes, and Marten Scheffer. 2012. Methods for detecting early warnings of critical transitions in time series illustrated using simulated ecological data. *PLOS ONE*, 7(7):e41010.

Elvis Dohmatob, Yunzhen Feng, and Julia Kempe. 2024a. Model collapse demystified: The case of regression. In *Advances in Neural Information Processing Systems 37 (NeurIPS)*.

Elvis Dohmatob, Yunzhen Feng, Arjun Subramonian, and Julia Kempe. 2025. Strong model collapse. In *The Thirteenth International Conference on Learning Representations (ICLR)*.

Elvis Dohmatob, Yunzhen Feng, Pu Yang, Francois Charton, and Julia Kempe. 2024b. A tale of tails: Model collapse as a change of scaling laws. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*.

Kawin Ethayarajh. 2019. How contextual are contextualized word representations? comparing the geometry of bert, elmo, and gpt-2 embeddings. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP-IJCNLP)*, pages 55–65.

Jun Gao, Di He, Xu Tan, Tao Qin, Liwei Wang, and Tie-Yan Liu. 2019. Representation degeneration problem in training natural language generation models. In *International Conference on Learning Representations (ICLR)*.

Quentin Garrido, Randall Balestriero, Laurent Najman, and Yann LeCun. 2023. RankMe: Assessing the downstream performance of pretrained self-supervised representations by their rank. In *Proceedings of the 40th International Conference on Machine Learning (ICML)*.

Matthias Gerstgrasser, Rylan Schaeffer, Apratim Dey, Rafael Rafailov, Henry Sleight, John Hughes, Tomasz Korbak, Rajashree Agrawal, Dhruv Pai, Andrey Gromov, Daniel A. Roberts, Diyi Yang, David L. Donoho, and Sanmi Koyejo. 2024. Is model collapse inevitable? breaking the curse of recursion by accumulating real and synthetic data. *arXiv preprint arXiv:2404.01413*.

Yi Gu, Lingyou Pang, Xiangkun Ye, Tianyu Wang, Jianyu Lin, Carey E. Priebe, and Alexander Aue. 2026. SIGMA: Scalable spectral insights for LLM model collapse. *arXiv preprint arXiv:2601.03385*.

Naicheng He, Kaicheng Guo, Arjun Prakash, Saket Tiwari, Ruo Yu Tao, Tyrone Serapio, Amy Greenwald, and George Konidaris. 2025. Spectral collapse drives loss of plasticity in deep continual learning. In *Advances in Neural Information Processing Systems 38 (NeurIPS)*.

Shayne Longpre, Gregory Yauney, Emily Reif, Katherine Lee, Adam Roberts, Barret Zoph, Denny Zhou, Jason Wei, Kevin Robinson, David Mimno, and Daphne Ippolito. 2024. A pretrainer's guide to training data: Measuring the effects of data age, domain coverage, quality, & toxicity. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*.

Charles H. Martin and Michael W. Mahoney. 2021. Implicit self-regularization in deep neural networks: Evidence from random matrix theory and implications for learning. *Journal of Machine Learning Research*, 22(165):1–73.

Niklas Muennighoff, Alexander M. Rush, Boaz Barak, Teven Le Scao, Nouamane Tazi, Aleksandra Piktus, Sampo Pyysalo, Thomas Wolf, and Colin Raffel. 2023. Scaling data-constrained language models. In *Advances in Neural Information Processing Systems 36 (NeurIPS)*.

Alireza Naderi, Thiziri Nait Saada, and Jared Tanner. 2025. Mind the gap: a spectral analysis of rank collapse and signal propagation in attention layers. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*.

Vardan Papyan, X.Y. Han, and David L. Donoho. 2020. Prevalence of neural collapse during the terminal phase of deep learning training. *Proceedings of the National Academy of Sciences*, 117(40):24652–24663.

Olivier Roy and Martin Vetterli. 2007. The effective rank: A measure of effective dimensionality. In *15th European Signal Processing Conference (EUSIPCO)*.

Marten Scheffer, Jordi Bascompte, William A. Brock, Victor Brovkin, Stephen R. Carpenter, Vasilis Dakos, Hermann Held, Egbert H. van Nes, Max Rietkerk, and George Sugihara. 2009. Early-warning signals for critical transitions. *Nature*, 461:53–59.

Marten Scheffer, Stephen R. Carpenter, Timothy M. Lenton, Jordi Bascompte, William Brock, Vasilis Dakos, Johan van de Koppel, Ingrid A. van de Leemput, Simon A. Levin, Egbert H. van Nes, Mercedes Pascual, and John Vandermeer. 2012. Anticipating critical transitions. *Science*, 338(6105):344–348.

Ilia Shumailov, Zakhar Shumaylov, Yiren Zhao, Nicolas Papernot, Ross Anderson, and Yarin Gal. 2024. Ai models collapse when trained on recursively generated data. *Nature*, 631:755–759.

Koen Siteur, Maarten B. Eppinga, Arjen Doelman, Enrico Siero, and Max Rietkerk. 2016. Ecosystems off track: Rate-induced critical transitions in ecological models. *Oikos*, 125(12):1689–1699.

Sang Michael Xie, Hieu Pham, Xuanyi Dong, Nan Du, Hanxiao Liu, Yifeng Lu, Percy Liang, Quoc V. Le, Tengyu Ma, and Adams Wei Yu. 2023. DoReMi: Optimizing data mixtures speeds up language model pretraining. In *Advances in Neural Information Processing Systems 36 (NeurIPS)*.

Jiasheng Ye, Peiju Liu, Tianxiang Sun, Jun Zhan, Yunhua Zhou, and Xipeng Qiu. 2025. Data mixing laws: Optimizing data mixtures by predicting language modeling performance. In *The Thirteenth International Conference on Learning Representations (ICLR)*.

A Per-Generation Cross-Seed Variance

Table 4 reports all six metrics at every generation under pure-synthetic training (r=0), with mean and sample standard deviation across three seeds (42, 43, 44). Spectral metrics (r_{eff}, SLV, σ_1) exhibit low cross-seed variance throughout the trajectory (SLV coefficient of variation $< 1.3\%$ at all generations), confirming that spectral collapse follows a deterministic geometric trajectory. In contrast, D-4 variance increases sharply after Gen 5, reaching a coefficient of variation of 46% at Gen 10, consistent with the stochastic nature of n-gram diversity under progressive mode collapse.

Table 4: Per-generation metrics under pure-synthetic training (r=0, Pythia-410M, mean$_{\pm \text{s.d.}}$ across seeds 42, 43, 44). Gen 0 shares a single pretrained checkpoint; D-4 variance at Gen 0 reflects generation sampling. Bold marks the spectral peak.

Gen	r_{eff}	SLV	PPL	HeSw	D-4	σ_1
0	$417.5_{\pm 0.0}$	$2366.7_{\pm 0.0}$	$9.55_{\pm 0.00}$	$.413_{\pm .000}$	$.738_{\pm .006}$	$1942.8_{\pm 0.0}$
1	$491.1_{\pm 0.6}$	$2462.3_{\pm 2.7}$	$11.37_{\pm 0.01}$	$.390_{\pm .001}$	$.609_{\pm .014}$	$808.9_{\pm 5.4}$
2	$\mathbf{496.2}_{\pm 0.3}$	$\mathbf{2468.0}_{\pm 1.7}$	$13.56_{\pm 0.02}$	$.380_{\pm .004}$	$.505_{\pm .011}$	$718.7_{\pm 7.0}$
3	$493.6_{\pm 0.4}$	$2440.4_{\pm 2.0}$	$16.23_{\pm 0.02}$	$.367_{\pm .005}$	$.465_{\pm .033}$	$708.7_{\pm 7.5}$
4	$487.8_{\pm 1.9}$	$2406.0_{\pm 2.0}$	$19.90_{\pm 0.09}$	$.354_{\pm .003}$	$.419_{\pm .047}$	$722.9_{\pm 6.9}$
5	$479.8_{\pm 0.5}$	$2366.0_{\pm 3.8}$	$24.50_{\pm 0.17}$	$.341_{\pm .003}$	$.412_{\pm .053}$	$749.2_{\pm 7.1}$
6	$472.4_{\pm 0.6}$	$2326.1_{\pm 8.6}$	$30.40_{\pm 0.45}$	$.329_{\pm .004}$	$.402_{\pm .067}$	$764.6_{\pm 5.7}$
7	$464.1_{\pm 0.2}$	$2293.2_{\pm 10.2}$	$37.82_{\pm 0.94}$	$.319_{\pm .000}$	$.375_{\pm .082}$	$788.4_{\pm 4.0}$
8	$455.8_{\pm 1.1}$	$2246.1_{\pm 7.4}$	$46.90_{\pm 2.41}$	$.311_{\pm .005}$	$.337_{\pm .137}$	$806.4_{\pm 8.0}$
9	$446.6_{\pm 2.1}$	$2210.4_{\pm 23.1}$	$58.56_{\pm 4.52}$	$.301_{\pm .003}$	$.316_{\pm .148}$	$841.1_{\pm 23.6}$
10	$438.1_{\pm 4.8}$	$2181.4_{\pm 28.3}$	$70.99_{\pm 5.56}$	$.296_{\pm .006}$	$.324_{\pm .150}$	$874.6_{\pm 44.2}$